



# Beets

## CULTIVARS

- Boro
- Chioggia
- Touchstone Gold
- Avalanche
- Red Ace (Pelleted)

## INTENSIVE SPACING

- 3 rows (11" - 30 cm) per bed spaced every 5" (12 cm) on the row
- Jang seeder 3X: LJ-12 13F-10B, brush 1-3 mm, ≠

## FERTILIZATION

- Beets do best in beds amended with well finished compost (rule of thumb is to spread compost ½ inch thick) but other amendments can supplement or replace it. Alternatively, we use a mix of alfalfa (2-0-2) meal and pelleted chicken manure (5-3-2).
- Calcium and Boron are important nutrients for beet and swiss chard. Without sufficient boron, beet growth suffers and the roots develop internal black spots.

## IMPLANTATION

We plant beets every 3 weeks to ensure a constant supply of fresh beets with nice greens. For every seeding, we plan enough beds for an estimated 4 weeks of harvest per planting.

## EQUIPMENT

- Rotary harrow
- 30" bed prep rake

- Direct Seeded: Individual Jang 3X seeder + LJ-12 roller, 5 markers for the rake, scale, notepad and your seeds
- Transplant: Popping tray and dibbler/marking roller

## **TRANSPLANT ITINERARY**

- Remove tarp and clean out any large debris. Use the rake if needed. \*
- Uniformly apply compost and or amendment and pass with the rotary harrow at a depth of 2" (5 cm). To make transplanting easier (seeing as the soil is loose after being harrowed), we only prepare the beds when we are about to transplant.
- Pass with the dibbler-roller to mark the planting holes and bring out the well-watered trays of transplants.
- Once the beds are ready, 2 people begin to drop seedlings while another person starts to plant. Use the popping tray to facilitate extraction of seedlings. 3 people/ 100' bed is good ratio for efficient workflow.
- While planting, pinch off extra heads if more than 1 seed has sprouted in the cell.
- Install irrigation and water until the soil is well moistened.

\* If the previous crop is too well established, or if the bed is too messy, mow the crop as low as possible and harrow at a depth of 2" (5 cm). Add 1" (2.5cm) of compost, evenly spread on the whole width of the bed and transplant into compost following a no-till strategy.

## **DIRECT SEEDING ITINERARY**

- Remove tarp and clean any large debris. Use the rake if needed.\*
- Uniformly apply compost and/or amendments and pass with the rotary harrow at a depth of 1.5" (4 cm). Use the rake to remove debris if needed. Otherwise, the bed is ready to be seeded.
- Mark 3 rows every 11" (28 cm) with the help of markers and seed with the Jang. Write down the total amount of seed dropped onto the first bed and compare with the target. Adjust calibration if required.
- Install irrigation and water until the soil is well moistened. Keep moist for the next 7-10 days until all seeds have germinated.

- Start some extra seedlings in the greenhouse to fill in the holes on the beds.

\*If the previous crop is too well established, or if the bed is too messy, mow the previous crop as low as possible, harrow the beds at a depth of 2" (5 cm) and rake it clean. Add 1" of compost on the bed and level well using the back of the rake to get the same thickness across the whole width of the bed. Seed directly onto compost following a no-till strategy.

## WEEDING

- Beets do not benefit from the canopy shading of high density spacing, and therefore need to be cultivated more than once to keep the crop clean.
- 10-14 days after transplant (20 days after direct seeding), use the bio disc to cultivate on the row and between the rows in one single pass.
- 7 to 10 days after initial pass with the bio-disc (or before small weeds reach the two leaf stage), use 5" stirrup hoe for a second cultivation.

## PLANT PROTECTION

- Boron and seaweed (or compost tea) is mix together and applied foliarly 20 days after transplanting (30 days after seeding). This is best done early in the morning in order to let the leaf absorb the foliar application. Avoid spraying during or before rain, as it will diminish the effect of the application.
- Cercospora leaf spot is controlled by wider spacing (3 rows, instead of 4) and proper crop rotation (wait at least 3 years before planting beets or swiss chard again on same beds). If still problematic, replace seaweed in the supplemental foliar spray by well brewed compost tea.

## HARVEST

- Beets are harvested with their tops and bunched in the fields in a kneeling position.
- Only the nice round beets are harvested and ALWAYS size the beets when bunching them, grouping those of the same size together. Leave in the fields any beets that have markings (black incisions) due to boron deficiency, or other significant defects. Take time to remove any rotten or yellow leaves.

- Baby beets should be from 1.5" to 2.5" (4 to 5 cm) in bunches of 5 to 6.
- For bigger beets, ranging from 2.5" to 3.5", put 3 to 4 per bunch.
- Make piles of 5 bunches in the pathways ready to be picked up in harvest crates at the end of the harvest.
- If you have a specific breakdown of how many bunches of each variety are to be harvested (ex. 60 chiogga, 70 golden, etc.), you can count out the number of elastic bands you have in your hand, prior to harvest, and then harvest that many bunches.
- 1 person should harvest at least 50 bunches per hour, depending on the size of the beets.
- When harvesting beets that won't be bunched, chop the greens off as you harvest those beets in the field.

## **CONDITIONING**

- Upon arriving from the fields, the harvest crates are placed to the left of the sinks. The bunches are then washed with a pressure gun one by one and piled on the right. Once the pile has reached a certain height, place bunches of the same cultivar in each storage bin and identify the bin.
- 1 person should wash and package at least 100 bunches per hour.

## **STORING**

- Beets are stored in cold room kept at 35°F (2°C)